



Effects of interventions using graphic organizer in Korea: a meta-analysis

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Abstract

This study aimed to evaluate the effects of intervention using graphic organizers on the cognitive and affective improvement of students with intellectual disability (ID), with learning disability (LD), without disability, and at-risk learners in Korea. A total of 49 peer-reviewed journals and dissertations for the last 20 years were included for conducting this meta-analysis. The overall effect size of intervention using a graphic organizer was .78 (d) (95% CI [.63, .94], $\tau^2 = .28$) using a random-effects model. In order of strongest to weakest effects, at-risk learners ($d = 1.38$), students with LD ($d = 1.15$), students with ID ($d = .76$), and students without disability ($d = .52$). Among student variables, there is no statistically significant difference by school level, but by school type. Among intervention variables, instruction in math ($d = 1.43$) and Korean ($d = .96$); cognitive mapping ($d = 1.05$); 1–19 times, 1–9 weeks, 3–5 times per week; and small-size groups were the most effective intervention conditions. While several variables showed significant subgroup differences, meta-regression analyses revealed that only group size and frequency were significant moderators after controlling for other factors. In summary, intervention using a graphic organizer was more effective for students with disability and at-risk students than it was for students without disability.

Keywords Graphic organizer · Students with disability · Meta-analysis

Effects of interventions using graphic organizer in Korea: a meta-analysis

Using graphic organizer (GO) is a widely applicable strategy for various types of learners across diverse school levels (Dexter & Hughes, 2011). The use of GO is not limited to general education settings; it is also implemented in special education settings and is considered an evidence-based practice for inclusive education (Kim et al., 2004). Most learners who were reported to benefit from GO were students in an inclusive education setting (Quenneville, 2001), and studies have been conducted to verify its effectiveness for students with and without disabilities, respectively (Dexter & Hughes, 2011; Kang, 2002; Kim et al., 2004). Given that few studies have compared the effects of GOs by learner type (Kim et al., 2013), a meta-analysis of the effectiveness of

GOs for various types of learners is meaningful. Hence, this study examined the moderator effects of GOs for all types of learners, with the aim of revealing the optimal conditions for effective GO use in an inclusive setting.

Theoretical background of graphic organizer

GO is a learning strategy to help learners connect their previous experiences or knowledge with new concepts (Dexter, 2010). Compared to text-only materials, the use of GOs is known to be effective in improving comprehension and knowledge acquisition (Moore & Readence, 1984). Originally, GO stemmed from Ausubel's (1960) theory of meaningful reception learning. The theory asserts that in order to acquire new knowledge, learners must undertake the process of relating new knowledge to their personal experiences and existing cognitive structures and in so doing approach understanding new concepts step-by-step (Johnson et al., 1997). Additionally, the core principle of GO is the acquisition of knowledge and concepts based on gradual differentiation through meaningful reception learning that specifically clarifies the relationship between abstract concepts (Ausubel,

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1960). In other words, the GO provides a schematic to produce changes in the cognitive structure of learning, and the learner uses a mental scaffolding strategy that connects new information with their personal experiences in order to acquire new knowledge (Dexter, 2010).

Research indicates that students who have difficulties in learning frequently encounter difficulties when it comes to linking new information with the previous knowledge, distinguishing between core and non-core contents, and implementing comprehension strategies (DiCecco & Gleason, 2002; Kim & Cha, 2010; Singleton et al., 2015; Vaughn & Edmonds, 2006). Thus, GO was considered as an effective strategy for students who have difficulties in learning including students with learning difficulties, disabilities, and with intellectual disabilities (Dexter et al., 2011; Kim et al., 2004). However, a few studies have attempted to examine the effectiveness when comparing different types of learners so far.

Empirical evidence supporting the effectiveness of graphic organizer

Evidence supporting the effectiveness of GO has been reported for various types of learners. For example, GO is effective at improving at-risk learners' reading skills (Gajria et al., 2007; Horton & Lovitt, 1989) as well as the writing outcomes of students with reading disabilities (Gillespie & Graham, 2014). Additionally, GO is an effective component of math interventions such as addition and subtraction operations among middle school students with a mathematical disability (Jitendra et al., 1999). GO has proven to be useful for enhancing counting and arithmetic skills among students with an intellectual disability (ID; Erez & Peled, 2001). Additionally, GO is useful for improving math and content subject area outcomes among middle school students with a learning disability (LD; Bulgren et al., 2007; Ives, 2007). In Korea, Kim et al. (2013) revealed the effectiveness of reading, writing, and math interventions using GOs for students with LDs and at-risk learners. In sum, per the empirical evidence, GO has been proven to be an effective intervention strategy for various types of learners.

However, to what extent the effectiveness of interventions using GOs differ by types of learners remains unknown. Several meta-analyses have been conducted involving specific types of learners, but these have been limited to either student without disabilities or student with LDs. For example, Joo (2019) conducted the first meta-analysis of GOs in Korea and excluded at-risk learners and students with disabilities; that is, the study only included students without disabilities. Furthermore, two other meta-analyses focused on students with LDs who showed less proficient language skills than students without learning difficulties (Dexter

et al., 2011) and students with IDs who showed a low literacy ability (Schwab, 2017), respectively. Since the early meta-analysis of GOs (Moore & Readence, 1984), there have been very few studies on the magnitude of GOs' effectiveness by type of learner.

Potential moderators of effectiveness on graphic organizer

The extant research on implementation of GO demonstrates its effectiveness as evidence-based practices (Dexter & Hughes, 2011; Dexter et al., 2011; Kim et al., 2004; Kim et al., 2013). Evidence-based practices are defined as practices "that are supported by multiple, high-quality studies that utilize research designs from which causality can be inferred and that demonstrate meaningful effects on student outcomes" (Cook & Cook, 2013, p. 73). Evidence-based practice is to not simply demonstrate the effectiveness of empirically validated instructional practices, but also to disseminate the understandings of whether its practices are effective for whom in a particular setting (Odom et al., 2005; Vaughn & Linda-Tompson, 2003). Though students are diagnosed with the same disability, the complexity and variability of each student's response to their environment differ (Horner et al., 2005). Therefore, it is crucial to identify the contexts in which GO works, while considering variables such as participant-related variables (e.g., school level) and intervention-related variables (e.g., intervention session, intervention frequency, group size, GO types).

School level

According to Gerber et al. (1995), the effectiveness of GO was affected by learners' developmental age due to cognitive process load for younger learners. In other words, younger learners may not understand the conventional meaning of graphics such as arrows (McTigue & Flowers, 2011; Roberts & Brugar, 2017). As evidence, Readence and Moore (1981) revealed that older learners benefit more from GO because they have less barriers to processing and integrating information than younger learners. In addition, Gerber et al. (1995) addressed that younger learners had difficulty perceiving graphics holistically, but in isolation. However, the recent meta-analysis by Guo et al. (2020) did not find statistically significant difference by effects of grade level, but still showed higher effect sizes for higher grades than lower grades. Though it is assumed that a higher cognitive process load is needed for benefiting from graphics and that scaffolding may be needed for younger learners (Schnotz, 2014), whether the effects of GO could be differed by

developmental age was not clarified enough in the previous studies.

In several prior meta-analyses, researchers indicated that the effectiveness of GO can be differed by students' school level. For example, Guo et al. (2020) and Luiten et al. (1980), synthesizing studies conducted in the United States, reported a small effect sizes derived from both the elementary and secondary levels. However, these findings were contradictory to the findings of Liu et al. (2014). A meta-analysis by Liu et al. (2014), including most of the studies conducted in China, reported large effect sizes derived from the elementary and high school and a small effect sizes derived from middle school. Therefore, it remains unclear on school level have moderator effects on intervention using GO. In addition, even the results of meta-analytic studies reported in English may not necessarily be generalizable to other cultural settings, such as Eastern education settings.

Intervention dosage: duration, frequency, and group size

According to taxonomy of intervention intensity suggested by Fuchs and Vaughn (2012), Fuchs et al. (2017), intervention dosage includes number of intervention sessions, frequency of intervention, and group sizes. As many meta-analyses contributed to examine the moderator effects of GO by intervention dosage, intervention dosage was considered as one of strong moderators that could lead significantly different effect sizes (Liu et al., 2014; Schroeder et al., 2018). Typically, researchers assumed that higher intervention dosage (higher number of intervention sessions, higher frequency of intervention, and smaller group sizes) could learners be more exposed to contents sufficiently, ultimately linking to their achievements (Donegan & Wanzek, 2021; Gersten et al., 2020; Jung et al., 2017; Wanzek & Vaughn, 2007). For example, Schroeder et al. (2018) explained that cognitive load did decrease when processing the formats of GO as learners are familiar with the formats and they become more likely to focus on the learning contents and materials. However, studies showed that higher intervention dosage does not always more correlated with higher effect sizes of interventions using GO. For example, Schroeder et al. (2018) revealed that implementing GO during less than one week even showed a moderate effect size. On the contrary, Liu et al. (2014) found that intervention durations of three to six months showed a higher effect size compared with those spanning one to two months or seven months and more. Nesbit and Adesope (2006) also revealed that shorter intervention duration is more associated with higher effects. Extant studies still showed inconsistent with the findings that the more intensive intervention provided in terms of

increasing numbers of sessions and frequencies, and smaller group sizes, the more effective for learners. Therefore, this current study aims to examine whether the effectiveness of GO is not always positively correlated with intervention dosage.

Types of graphic organizer

In several meta-analysis of GO, researchers have examined the moderator effects sizes by types of GO (Dexter & Hughes, 2011; Dexter et al., 2011; Kim et al., 2004; Kim et al., 2013). According to Kim et al. (2013), the types of GO are classified into visual display, cognitive mapping, semantic mapping, and story maps. In the most of previous meta-analysis, it is not clear which type of GO could be more effective than the others. For example, Guo et al. (2020), the recent meta-analysis, revealed that single type of GO does not showed significantly different each other, but picture GO is significantly more effective than mixed GO. This study provided the implications for future study in that using specific GO only can be more effective than mixed GO. Thus, this current study aims to compare the effects of each GO only (cognitive mapping, semantic mapping, and story prompt) can be more effective than mixed GO using more than one type of GO.

Outcome types

Extant meta-analysis studies have demonstrated the effectiveness of GO in cognitive and affective outcome (Dexter et al., 2011; Joo, 2019; Kim et al., 2004). Cognitive outcome refers to learners' academic performance such as achievements in specific subject, cognitive ability, strategy-using ability (Joo, 2019; Kim et al., 2013; Liu et al., 2014). Affective outcome refers to learners' attitude and interest on the subject, and self-efficacy (Joo, 2019; Kang, 2004; Liu et al., 2014). Liu et al. (2014) revealed the difference of effect sizes of GO on cognitive and affective outcomes. They found that GO is more effective on cognitive outcome than affective outcome, though they did not compare the statistically significant difference between two outcomes. However, many meta-analysis studies not only still reported the aggregated effect sizes of cognitive and affective outcomes (Guo et al., 2020), but also other studies focused on only one aspect of outcome types (Dexter et al., 2011; Kim et al., 2013). Therefore, this current study aims to examine the statistical difference between two types of outcomes.

Purpose of the study

The main reason for the lack of meta-analyses of GOs in Korea is the insufficient number of accumulated studies (Kang, 2004; Kim et al., 2013; Oh & Nam, 2015). However, a sufficient number of experimental design studies examining the effectiveness of GOs have been conducted recently. Therefore, the current meta-analytic study sought to synthesize the effectiveness of interventions using GOs across types of learners. This study also aimed to examine the moderator effects of interventions using GOs by student variables, intervention variables, study design, publication status, and dependent variables. As mentioned, previous meta-analyses conducted in different countries have reported inconsistent findings with respect to moderator variables, and the results cannot be generalized to all settings. This study aims to encourage researchers abroad to compare the results obtained in Korea to those obtained in their own country settings. To this end, this study addresses the following research questions:

Research Question 1: What is the magnitude of the overall effect of interventions using GOs by types of learners (students without disabilities, students with LDs, students with IDs, and at-risk learners)?

Research Question 2: Do the effects of GO differed by subgroup of variables (student, intervention, and dependent) on outcomes for all learners?

Research Question 3: Are the effects of GO moderated by school level, intervention dosage, GO types (single GO versus mixed GO), and outcome types for all learners?

Method

Data collection

Data searches

In this study, the author searched and collected dissertation and peer-reviewed articles conducted from 2000 to 2020 in Korea using the following major domestic databases: the Academic Research Information Service, the Institute of Academic Information, and the National Assembly Electronic Library. The author searched for articles using the following keywords terms through the databases: graphic organizer, mapping, diagram, and schema.

The inclusion criteria are as follows. First, studies had to include students in elementary and secondary schools. Second, studies had to include learner types such as students without difficulties in learning, at-risk learners, students with LD, and students with ID. Third, only studies that implemented the intervention using GO for academic

skills including reading, writing, math, and other academic areas were included. Fourth, if study design includes control groups, it should be a group that receives intervention business as usual. Fifth, experimental and quasi-experimental studies that conducted single-group post-test design, single-group pre- and post-test design, and two-group pre- and post-test design were included. We decided to include single-group pre-post-design to this current meta-analysis. This is because we believed the benefits of comprehensively synthesizing the effects of GO, including single-group pre-post-design, outweighs the methodological concerns (e.g., Cheng et al., 2023; van de Wetering et al., 2022).

Study selection

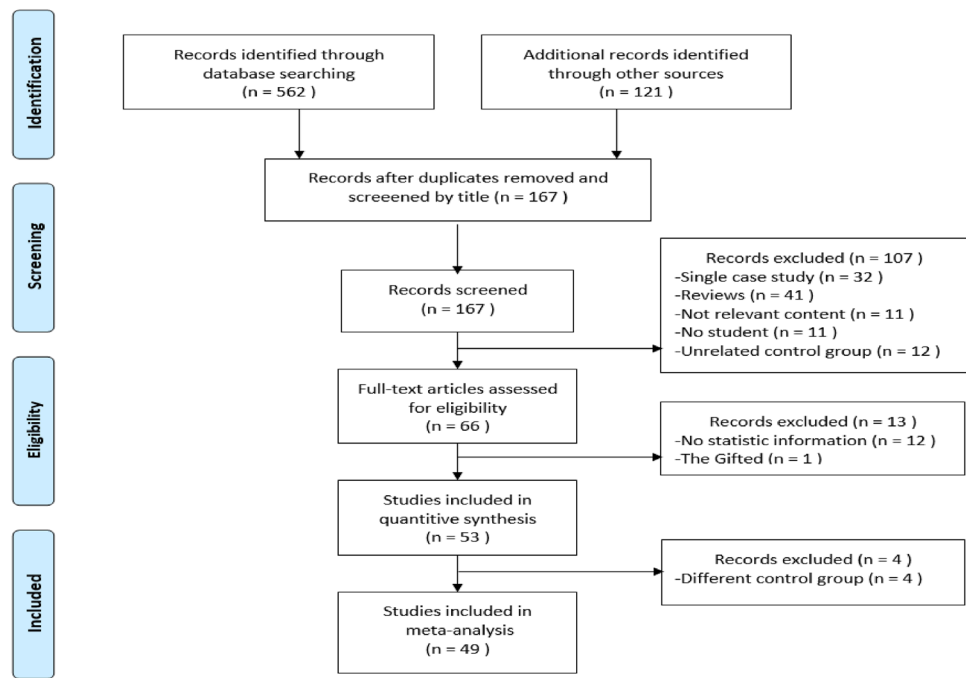
After searching, the number of articles identified was 562, and the number of additional articles included after reviewing the reference section of relevant studies was 121. After removing duplicates and eliminating the irrelevant studies by title, the number of remaining documents was 167. While assessing the full-text of articles, the author excluded 32 single case experimental design studies, 11 not relevant content studies with intervention using GO, 41 literature review studies, 11 studies that did not include intervention for students, and 12 studies in which control groups were not business as usual, for a total of 107 studies. After assessing full-text, a total of 66 articles remained. The number of studies included in the quantitative analysis was 53, excluding 12 studies for which the effect size could not be calculated due to lack of statistical values and one study that focused on gifted learners, which did not meet the criteria. Four studies were excluded that have control group either including different learner types or using other strategies rather than intervention business as usual. Finally, a total of 49 studies were included in the meta-analysis. All of the procedures from identification, screening, eligibility, to inclusion followed the PRISMA flow chart (Moher et al., 2009), and procedures for excluding articles are described in Fig. 1.

Data coding

Coding variables

The author coded study characteristics, student, intervention, study design, publication status, and dependent variables. Study characteristics include the title, authors, and publication year. Student variables include placement, school level, and learner type. Placement includes general class, special class, and self-contained classroom in a general school. School level includes elementary and secondary school. Learner type includes students with ID, students with LD, students without difficulties in learning, and at-risk learners. Students without difficulties in learning were defined as

Fig. 1 PRISMA flow chart



students who were not reported to have any significant difficulties in learning academic instruction in studies examined (An, 2020; An et al., 2023). At-risk learners are defined as learners who were reported to have low achievement and struggling in learning, requiring supplementary intensive intervention in studies examined (An et al., 2023; Kim et al., 2022). Students with LD are defined as students who do not respond to supplementary intensive intervention for at least a semester and require individualized intervention (Vaughn and Linan-Thompson (2003). The rate of students with LD were reported typically about 5–10% among total of school year students across countries (Fuchs & Vaughn, 2012). Students with ID are defined as students who have significant limitations in intellectual functioning and adaptive behaviour, characterized by an IQ score below 70, and difficulties in areas such as social skills, communication, self-care, and independent living (Kim et al., 2020). If a control group exists in the study, all types of learners in the treatment group and control group in each study should be same.

Intervention variables were classified into subject, intervention area, GO type, session, duration, frequency, and group size. Subjects include Korean, English, math, social studies, science, computer, and mixed across multiple subjects. Intervention area includes reading, writing, and listening. Graphic organizer type includes visual display, semantic mapping, cognitive mapping, and story mapping (Dexter & Hughes, 2011; Kim et al., 2013). Intervention dosage includes sessions, duration, and frequency. We considered prior research on intervention dosage when categorizing each intervention dosage variable (Donegan

& Wanzek, 2021; Gersten et al., 2020; Jung et al., 2017; Wanzek & Vaughn, 2007). Total number of intervention sessions includes one to nineteen and more than nineteen sessions. Intervention duration includes one to nine and more than nine weeks (Kim et al., 2022). Intervention frequency includes one to two sessions per week and three to five sessions per week. Group size includes small (one to five students) and large group size (more than five students).

Study design was coded among a single-group pre-post, two-group post, and two-group pre-post. Publication status was coded in either dissertations or peer-review articles. Dependent variables included cognitive and affective domains. Cognitive domain included academic achievement, cognitive ability, self-directed ability in cognitive skills, and learning strategy. Affective domain included attitude, interest, and self-efficacy (Kang, 2002).

Reliability of studies selection and data coding

The author selected following the data search and inclusion criteria. Two special education doctoral students independently checked whether the randomly selected 20% of studies was consistently conducted in the stages of screening and assessing the full-text. The interrater reliability in each stage reached 100%.

The author coded the variables of final studies included. Then, the author provided training to two coders, doctoral students in special education, with manual materials of the coding framework to check whether the randomly selected

20% of studies were accurately coded. The interrater reliability was calculated by numbers of agreements divided by numbers of agreements and disagreements. The interrater reliability was 96% on average. Discrepancies between were resolved after discussion between three coders.

Data analysis

Calculating effect sizes

Effect size is an index representing the size of effectiveness of the intervention. The magnitude of the Cohens' d effect size (d) can be classified as small ($.2 < d < .5$), medium ($.5 < d < .8$), or large ($d > .8$) (Cohen, 2016). The author used the Cohens' d formula proposed by Borenstein et al. (2011) to calculate the experimental studies such as single-group pre- and post-test, two-group post-test, and two-group pre- and post-test design. The author calculated individual effect size and used 'metagen' function to aggregate the overall effect size and examine the subgroup effect size through the 'meta' and 'metafor' package of the R Studio 1.3.959. We also conducted the meta-regression analysis using 'robumeta' package in R studio. Correlation coefficient (r) between moderators ranged from |.01| to |.38|, indicating a low risk of multi-collinearity between moderators (Appendix A).

Single group pre and post-test

Cohen's d was calculated as the mean difference of pre- and post-score by treatment group ($\overline{X_{diff}}$) divided by pooled standard deviation (S_p). Correlation (r) was pre-post-correlation rates and it was coded as .50, because there was no correlation information reported by precedent studies (Borenstein et al., 2011).

$$\overline{X_{diff}} = \overline{X_{post}} - \overline{X_{pre}}, d = \frac{\overline{X_{post}} - \overline{X_{pre}}}{S_p},$$

$$S_p = \sqrt{S_{pre}^2 + S_{post}^2 - (2 * r * S_{pre} * S_{post})}$$

Two group post-test

n_1 Is the number of control group cases, n_2 is the number of experimental group cases. S_p is pooled standard deviation within the treatment and control group. Cohen's d is calculated by subtracting the mean difference of post-score by treatment and control group (MD), then dividing by the pooled standard deviation (S_p).

$$MD = \overline{X_1} - \overline{X_2}, SMD = \frac{\overline{X_1} - \overline{X_2}}{S_p},$$

$$S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 + n_2 - 2)}}$$

Two group pre and post-test

n_c Is the number of control group cases, n_t is the number of experimental group cases. S_p is the pooled standard deviation within the treatment and control group. Cohen's d is calculated by subtracting the mean difference of pre-post-score of treatment group ($\overline{D_T}$) by the mean difference of pre-post of the control group ($\overline{D_C}$), then dividing by the pooled standard deviation (S_p).

$$MD = \overline{D_T} - \overline{D_C}, SMD = \frac{\overline{D_T} - \overline{D_C}}{S_p},$$

$$S_p = \sqrt{\frac{(n_1 - 1)S_T^2 + (n_2 - 1)S_C^2}{(n_1 + n_2 - 2)}}$$

Heterogeneity test

Heterogeneity can be examined visually through a forest plot identifying the difference of mean ES between studies. The degree of overlap of the confidence intervals (CI) of the effect sizes presented in the forest plot would be identified (Borenstein et al., 2011). The I -squared homogeneity test is also used to identify by a statistical test of heterogeneity (Higgins et al., 2003). If the I -squared estimate of homogeneity test is 25% or more and less than 50%, it is small; if it is 50% or more and less than 75%, it is medium; and if it is 75% or more, it is considered to be large extent of heterogeneity between studies (Higgins et al., 2003).

Random-effects model

It is important for the researcher to decide between the fixed-effect model and the random-effect model based on the decision that the characteristics of the study group are homogeneous or heterogeneous (Başol, 2016; Borenstein et al., 2011; Rothstein et al., 2013). After considering the different natures and settings of the participants included across studies and the main inference that we made, which is to identify subgroup differences (Hedges & Vevea, 1998; Konstantopoulos & Hedges, 2009), researchers have decided to adopt a random-effects model.

Dependence

This meta-analysis included 237 individual effect sizes from 49 studies. With multivariate variables in a study, there is dependence within the same study. Therefore, to prevent violating the dependence of effect size, the authors used weighted random-effects robust variance estimation (Hedges et al., 2010) using *robumeta* package (Fisher & Tipton, 2015) to conduct subset analysis and meta-regression analysis. The robust variance estimation considers the assumptions that individual effect sizes are nested within samples of the study, requiring the mean correlation between all the pairs of effect sizes within samples (ρ ; ρ). We chose for a within-study correlation level of $\rho = .80$ (Filderman et al., 2022; Jitendra et al., 2021; Shanahan et al., 2024) using this as mean correlation between all the pairs of effect sizes within a study. We found that the variation in ρ values ($\rho = .80, .50$, and $.20$) did not induce any changes in g or its standard error (Appendix B). The authors incorporated a small-sample bias correction ($\text{vcov} = \text{"CR2"}$ was coded in *robu* function), adjusting for both degrees of freedom and residuals during coefficient estimation to prevent Type 1 error (Tipton, 2015).

Sensitivity analysis

The authors identified eight articles of low-quality as assessed by the essential quality indicators of Gersten et al. (2005) using 10 main questions, including conceptualization, participant, implementation of intervention, outcome measures, and data analysis. After evaluating studies using the quality indicators the author identified a total of eleven low-quality studies that met less than 50% of quality indicators among 10 questions.

Results

Descriptions of studies

Appendix C provides the study characteristics and effect size for the 49 studies. In seven studies, the learner type was at-risk learners, eight focused on students with LD, six focused on students with ID, 28 focused on students without disability. In 31 studies, school level was elementary, 18 studies were in secondary school. In 37 studies, placement was general class, 11 were in self-contained classrooms in the general school, and one was in a special school. Thirty-seven studies employed two-group pre-post-test, nine studies employed two-group post-test, and three studies employed single-group pre- and post-test designs. Twenty-six studies were peer-reviewed studies and 23 studies were dissertations. Thirty-two studies assessed cognitive outcomes and

seventeen studies assessed both cognitive and affective outcomes.

Sensitivity analysis

Three single-group pre- and post-studies and eight two-group post studies were identified as potential low quality of studies that required a sensitivity analysis to determine whether these designs could impact the overall effect size. The author conducted the sensitivity analysis to compare the effect sizes, confidence interval, and heterogeneity between inclusion and removal of eleven low-quality articles. We did not find a significant difference in interpreting the overall effects sizes and confidence intervals results between the inclusion model ($d = .78$, 95% CI [.63, .94], $I^2 = 45.7\%$, $\tau^2 = .28$) and the removal model ($d = .83$, 95% CI [.63, 1.02], $I^2 = 54.2\%$, $\tau^2 = .39$).

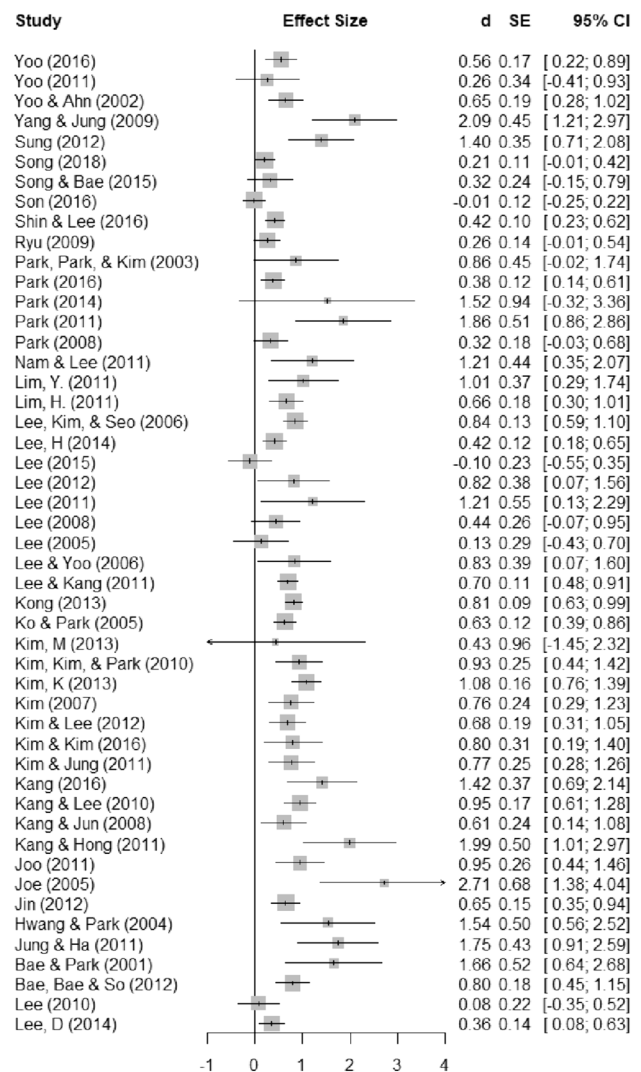


Fig. 2 Results of effect sizes by studies

Therefore, the author included all of the articles to conduct this meta-analysis.

Heterogeneity test for random effect model

To assess heterogeneity, the author identified the results that I -squared is more than 50% and the homogeneity verification probability (p) is less than .10 (Higgins et al., 2003), and the heterogeneity was confirmed in the forest plot in Fig. 2. Based on these results, the author concluded there was empirical evidence that supported the heterogeneity of effect size between the studies ($Q=109.18$, $df=48$, $p<.0001$, I -squared=56%), and the author adopted the random-effect model in this meta-analysis. In addition, it is reasonable to consider the random-effect model in that different studies including various learner types, especially students with disabilities are heterogeneous (Odom et al., 2005). The heterogeneity hypothesis was accepted, the random-effects model was adopted and this model estimates the distribution of effect sizes considering a sampling error and the degree of disparity from the mean effects between studies.

Overall effect size and subset analysis

The overall effect size of intervention using GO was $d=.78$ and showed middle-sized effect size ($p<.001$, 95% CI [.63, .94], $\tau^2=.28$), and it ranged from $-.10$ to 2.83 (d). Next, Subset analyses for were conducted by student, intervention, and dependent variables (Table 1). Results of the meta-analysis and the test of heterogeneity showed that the effects of interventions using GO were differentiated by learner types. By learner types, the intervention using GO for at-risk learners ($d=1.38$, $p=.052$, $\tau^2=.09$), students with LD ($d=1.15$, $p<.05$, $\tau^2=.75$), and students with ID ($d=.76$, $p<.05$, $\tau^2=.04$) showed large-sized effect size. However, intervention using GO for students without disability ($d=.52$, $p<.001$, $\tau^2=.01$) showed a middle-sized effect size.

By school level, the effect size of intervention using GO in elementary school was $d=.82$ (95% CI [.59, .1.05], $\tau^2=.33$) and that of intervention using GO in secondary school was $d=.60$ (95% CI [.45, .75], $\tau^2=.01$), both with significant positive effects ($p<.001$). By placement, studies conducted in the self-contained classroom in the general education class showed a large-sized effect size ($d=.83$, 95% CI [.62, 1.04], $\tau^2=0$), studies conducted in the general class showed a middle-sized effect size ($d=.74$, 95% CI [.55, .92], $\tau^2=.36$).

A category of special education school was excluded from this analysis because only one study included this category, and it did not meet the minimum quantity required for conducting subset analysis.

By subject area, the effect size of math intervention was $d=1.43$ (95% CI [.13, 2.72], $\tau^2=2.21$), but it did not show

a significant effect ($p=.106$). The effect size of Korean was $d=.96$ (95% CI [.65, 1.27], $\tau^2=.34$) and the effect size of English was $d=.48$ (95% CI [.31, .64], $\tau^2=.10$), both with significant positive effects ($p<.001$). Categories of computer, science, social studies, and mixed subjects were excluded from this analysis because few studies included this category, and it did not meet the minimum quantity required for conducting subset analysis.

By graphic organizer type, cognitive mapping approach showed a large-sized effect size ($d=1.05$, $p<.01$, 95% CI [.61, 1.48], $\tau^2=1.32$), whereas mixed GO use ($d=.71$, $p<.001$, 95% CI [.52, .90], $\tau^2=.17$) and semantic mapping ($d=.65$, $p=.053$, 95% CI [.25, 1.04], $\tau^2=.01$), and story mapping ($d=.58$, $p<.05$, 95% CI [.28, .87], $\tau^2=.08$) showed middle-sized effect size. Visual display was excluded for the moderator analysis since it did not meet the minimum quantity number for conducting subset analysis.

By the number of session, sessions from one to nineteen times showed the effect size of $d=.77$ (95% CI [.58, .95], $\tau^2=.26$) and sessions more than nineteen times showed the effect size of $d=.67$ (95% CI [.53, .80], $\tau^2=.26$), both with significant positive effects ($p<.001$). By duration, duration from one to nine weeks showed the effect size of $d=.79$ (95% CI [.58, 1.01], $\tau^2=.21$) and duration more than nine weeks showed the effect size of $d=.59$ (95% CI [.39, .78], $\tau^2=.14$), both with significant positive effects ($p<.001$). By frequency, more than two times per week showed the effect size of $d=.64$ (95% CI [.44, .84], $\tau^2=.20$) and one to two times per week showed the effect size of $d=.94$ (95% CI [.73, 1.14], $\tau^2=.12$), both with significant positive effects ($p<.001$). By group size, the group consisting of one to five students in a group showed the effect size of $d=1.58$ ($p=.054$, 95% CI [.73, 1.14], $\tau^2=.12$) and the group consisting of more than five students in a group showed the effect size of $d=.94$ ($p<.001$, 95% CI [.73, 1.14], $\tau^2=.12$).

By cognitive outcome, achievement showed the effect size of $d=.60$ (95% CI [.43, .77], $\tau^2=0$), cognitive ability showed the effect size of $d=.73$ (95% CI [.55, .91], $\tau^2=.11$), and strategy using ability showed the effect size of $d=.94$ (95% CI [-.19, 2.08], $\tau^2=7.44$). By affective outcome, attitude showed the effect size of $d=.61$ (95% CI [.07, 1.14], $\tau^2=.14$), interest showed the effect size of $d=.59$ (95% CI [.35, .84], $\tau^2=.27$), and self-efficacy showed the effect size of $d=1.01$ (95% CI [.39, 1.62], $\tau^2=1.09$).

Meta-regression analysis

We examined whether the overall effect size of GO is moderated by school level, intervention dosage, GO types (single GO versus mixed GO), and outcome types for all learners. We revealed that frequency (one to two times a week versus more than two times a week) and group size (one to

Table 1 Subset analysis of participant, intervention, dependent variables

	<i>N</i>	<i>k</i>	<i>df</i>	<i>d</i>	95% CI	Between-study sampling variance (τ^2)
Learner type						
Student with ID	6	25	2.89	0.76*	[0.30–1.22]	0.04
Students with LD	8	45	3.66	1.15*	[0.53–1.76]	0.75
Students without disability	28	143	17.1	0.52***	[0.40–0.64]	0.01
At-risk students	7	23	2.57	1.38	[1.06–2.16]	0.09
School level						
Elementary	31	128	16.5	0.82***	[0.59–1.05]	0.33
Secondary	18	109	10.4	0.60***	[0.45–0.75]	0.01
Placement						
Self-contained classroom in a general school	11	40	4.55	0.83***	[0.62–1.04]	0
General class	37	184	20.3	0.74***	[0.55–0.92]	0.36
Subject						
Korean	22	104	10.7	0.96***	[0.65–1.27]	0.34
English	16	94	10.2	0.48***	[0.31–0.64]	0.10
Math	6	8	3.53	1.43	[0.13–2.72]	2.21
Graphic organizer type						
Cognitive mapping	7	18	4.44	1.05**	[0.61–1.48]	1.32
Semantic mapping	6	30	2.73	0.65	[0.25–1.04]	0.01
Story mapping	7	31	3.34	0.58*	[0.28–0.87]	0.08
Mix	28	157	16.7	0.71***	[0.52–0.90]	0.17
Session						
1–19	36	173	18.7	0.77***	[0.58–0.95]	0.26
More than 19	14	52	8.23	0.67***	[0.53–0.80]	0
Duration (week)						
1–9	34	145	17.1	0.79***	[0.58–1.01]	0.21
More than 9	13	73	8.19	0.59***	[0.39–0.78]	0.14
Frequency (per week)						
1–2	29	157	16.5	0.64***	[0.44–0.84]	0.20
3–5	18	58	9.71	0.94***	[0.73–1.14]	0.12
Group size						
1–5	2	8	1	1.58	[1.31–1.84]	0
More than 5	47	229	26.2	0.70***	[0.56–0.85]	0.19
Cognitive outcome						
Achievement	9	11	6.91	0.60***	[0.43–0.77]	0
Cognitive ability	42	173	20.1	0.73***	[0.55–0.91]	0.11
Strategy using ability	2	5	1	0.94	[–0.19–2.08]	7.44
Affective outcome						
Attitude	8	11	5.05	0.61	[0.07–1.14]	0.14
Interest	13	22	9.76	0.59	[0.35–0.84]	0.27
Self-efficacy	7	14	4.48	1.01	[0.39–1.62]	1.09

Robust variance estimation is reflected to effect size and standard error. The total number of studies were 49, but several studies did not include information by moderators or excluded in this analysis due to small number of effect sizes. For example, a study (Shin and Lee (2016), that was conducted in special education school were excluded in the subset analysis by placement

k = number of effect sizes, *N* = number of studies

p* < .05, *p* < .01, ****p* < .001

five students in a group versus more than five students in a group) are statistically significant moderators when controlling for other factors, such as school level (secondary school versus elementary school), the number of sessions (more than 19 sessions versus 1–19 sessions), duration (more than 9 weeks versus 1–9 weeks), GO type (single GO type versus mixed GO), and outcome type (cognitive versus affective outcomes). The results of meta-regression analysis revealed that smaller group sizes were associated with significantly higher effect sizes than larger group sizes ($\beta = -.64, p < .05$). In addition, interventions provided more than two times per week were associated with significantly higher effect sizes than interventions provided one to two times per week ($\beta = .40, p < .05$). In contrast, school level, session, duration, GO type, and outcome types were not significant moderators when each moderator was controlled for in the meta-regression analysis. We present the results of meta-regression analysis in Table 2.

Independent analysis of different control groups

In this meta-analysis, studies in which the control group of business as usual without GO were included. The author

additionally examined six studies that conducted intervention to compare different conditions (see Table 3). Park (2012) reported that use of GO for students with LD showed a large-sized effect size compared with students without disability ($d = 1.90$). Park (2012) also reported intervention using GO and writing procedure strategy together was less effective than intervention using writing procedure strategy alone ($d = -.10$). Jun and Park (2001) reported intervention with student-constructed GO showed a large-sized effect size compared with teacher-constructed GO ($d = 1.95$). In addition, Kim (2003) reported that student-constructed GO showed a smaller effect size than teacher-constructed GO ($d = .14$). This result contradicted the result of Heo (2013) that reported that expert-constructed GO was slightly more effective than student-constructed GO ($d = .20$). Song and Bae (2015) reported employing the student-constructed organizer which is less structured and can be used more flexibly based on students' need, was more effective than employing the teacher-constructed organizer which is more structured and implemented in a standardized way ($d = .61$). Yoo and Na (2015) reported intervention that presented the GO at the beginning of class was more effective than presented the latter half of class ($d = .77$).

Publication bias

Publication bias is the tendency that only studies with large effect sizes are reported or peer-reviewed articles (Borenstein et al., 2011; Clarke & Clarke, 2000; Dickersin, 2005; Hedges & Vevea, 1998; Rothstein et al., 2013). We could check the trim-and-fill results which are visually symmetric presented in the funnel plot in Appendix D, there is no empirical evidence supporting results of the Egger's test ($p = .588, t = .55, df = 65$). The value of bias was .36 ($SE = .66$). The publication bias cannot readily identify only with funnel plot (Stern & Egger, 2001), but it is needed to see whether there is statistical evidence to support the slope change before and after trim-and-fill (Duval & Tweedie, 2000). Thus, from the results of funnel plot and Egger's test,

Table 2 Meta-regression moderator analysis

Moderator (baseline group)	Estimate	SE	<i>t</i>	<i>df</i>
Intercept	1.43	0.162	8.89***	15.88
School level (elementary)	− 0.13	0.196	− 0.68	12.93
Session (1–19 sessions)	− 0.20	0.165	− 1.22	10.80
Duration (1–9 weeks)	− 0.14	0.218	2.72	13.26
Frequency (1–2)	0.40	0.147	2.79*	13.26
Group size (1–5)	− 0.64	− 0.263	8.32*	8.32
GO type (mix)	− 0.06	0.200	− 0.30	14.90
Outcome type (affective)	0.03	0.157	0.23	15.86

Robust variance estimation is reflected to effect sizes and standard errors

SE standard error

* $p < .05$, ** $p < .01$, *** $p < .001$

Table 3 Independent analysis for other conditions of control group

Study name	Experimental group	Control group	<i>d</i>	− 95% CI	<i>SE</i>	+ 95% CI
Park (2012)	Students with LD	Students without disability	1.90	0.01	0.96	3.79
	Organizer + writing procedure strategy	Writing procedure strategy only	− 0.10	− 1.93	0.93	1.73
Jung and Park (2001)	Student-constructed	Teacher-constructed	1.95	− 1.15	1.58	5.05
Kim (2003)	Student-constructed	Teacher-constructed	0.14	− 0.82	0.49	1.11
Heo (2013)	Expert-constructed	Student-constructed	0.20	− 0.27	0.24	0.66
Song and Bae (2015)	Teacher-constructed	Student-constructed	0.61	0.05	0.28	1.16
Yoo and Na (2015)	Presented early	Presented later	0.77	0.36	0.21	1.18

the author concluded that there was no potential publication bias that could affect the overall interpretation of this study.

Sensitive analysis

A potential outlier were studies of Joe (2005) with effect size of 2.83, Lee (2011) with effect size of 2.48, and Yang and Jung (2009) based on visual analysis of the forest plot in Fig. 2. The authors conducted a sensitivity analysis to determine whether the inclusion and exclusion of this study affects interpretation of the overall effect size of the meta-analysis. After excluding the Joe (2005), Lee (2011), and Yang and Jung (2009), the overall effect size was $d = .72$ (95% CI [.58, .85], which is still the medium-sized effect size and inclusion of this study did not differ significantly from the exclusion model. In addition, the heterogeneity was acceptable; thus we decided this inclusion model allows more appropriate interpretation in practice ($p < .001$, $d = .78$, 95% CI [.63, .94], $I^2 = 45.7$).

Discussion

The purpose of this study was to evaluate the effects of interventions using GOs among elementary and secondary school students and compare effect sizes across each type of learner represented within the sample of students (i.e., those with IDs, those with LDs, those without disabilities, and at-risk learners). This meta-analysis included 49 studies in the final analysis; the objective was to inform decision making regarding evidence-based interventions using GOs. First, overall effect sizes were compared across each type of learner. Second, this study tested subgroup difference by student variables, intervention variables, study design, publication status, and dependent variables using a random-effects model. Third, we conducted meta-regression to examine whether overall effects of GO moderated by school level, intervention dosage, GO types (single GO versus mixed GO), and outcome types.

Overall effect and effect by learner type

The overall effect size was moderate at .78 (d); however, it closely approached a large effect size ($d > .8$). On the one hand, the overall effect size in this study was considerably higher than that of Schroeder's (2018) previous meta-analysis that synthesized the effect of concept mapping and had an effect size of .54 (d) as well as that of Guo (2020) (Hedges' $g = .39$). On the other hand, this study's overall effect size was considerably lower than that of Dexter and Hughes' (2011; $d = .91$). This result revealed that interventions using GOs conducted in Korea have shown a relatively large effect size.

Given that effect sizes varied across types of learners, specifically, it was observed that at-risk learners responded the most effectively, followed by students with LDs, students with IDs, and students without disabilities; these results indicated that the effect of the GO is relatively stronger for students who struggle in academic learning. Providing visual materials allows students with disabilities and at-risk learners to recognize relationships between abstract concepts (Dexter, 2010). It was found that using learning strategies and cognitive scaffolding through the GO is effective for students with low academic achievement (Ausubel, 1960). Kim (2013a, 2013b) provided support for these results, indicating that the GO is effective in learning strategies for students with disabilities and at-risk learners. Kang (2004) explained this by suggesting that GOs work in the role of schemas as a means of processing existing meanings and constructing new meanings. This contributes the resolution of the information processing and composition problems that students with disabilities and at-risk learner's encounter.

Subset analyses

The results for effect sizes by student variables revealed that elementary school students showed a large effect size, whereas secondary students showed a moderate effect size. This result suggests that the GO is relatively more effective for students at lower school levels, which aligns with previous studies (Guo, 2020; Kansızoğlu, 2017; Liu et al., 2014; Moore & Readence, 1984). These results may be explained by considering that most interventions using GOs in Korea tend to be effective, especially regarding students' basic academic learning in lower grades, such as reading, writing, and math (Joo, 2019; Kang, 2004).

Given that the effect sizes for Korean language and mathematics showed high effect sizes, it is suggested that the effect of the GO is stronger in interventions for acquiring basic learning abilities than interventions aimed at acquiring content knowledge and concepts (Dexter, 2010). Additionally, the effect of the GO has been found to be stronger among students learning Korean than among those learning English. On the one hand, for Korean students whose first language is Korean, the most frequent intervention area comprises basic learning abilities such as reading, writing, listening, and vocabulary. On the other hand, English is a second language for Korean students. Thus, the difference in the effects of using GOs between Korean and English might arise from students' difficulties in acquiring a foreign language compared to the relative ease of using their native language. Even at-risk students and students with disabilities who are vulnerable regarding basic academic skills may not respond to English language learning. However, these results contradict those of Kansızoğlu (2017) who found that the effect of the GO showed a large effect size as a component

of interventions targeting both English and native language learning. Presently, Korea still does not have a large body of studies comparing interventions using GOs for native and second languages.

The effect sizes observed according to the types of GOs were higher in order of cognitive mapping ($d = 1.05$), semantic mapping ($d = .65$), and story mapping ($d = .58$). This result contradicts that of Dexter (2011), for whom semantic mapping showed the highest effect size at 1.25 (d). In most cases, studies used multiple types of GOs simultaneously. Among the studies that only used one type of GO, semantic mapping and story mapping were the most frequently used. This is because many studies did not use consensual categories for the types of GOs. For example, Guo (2020) classified the types of GOs as pictorial diagram, picture, flow diagram, and mixed. Moreover, Oloyede (2011) compared the pictorial organizer with the written organizer and concluded that the former is stronger. Additionally, GO type classification has not been consistent across other studies (Dexter & Hughes, 2011; Kansizoğlu, 2017; Kim et al., 2004). Thus far, few studies have compared the GO's effectiveness across the various types of GOs; most studies have examined multiple types of GOs together. It would be meaningful to explore whether the different moderator variables can interact with each other, as this would shed light on whether certain types of GOs are more suitable in certain subject areas.

The strategy-using ability showed the highest effect size in the cognitive domain. This result aligns with that of Kang (2002) who reported the highest effect size for strategy knowledge, followed by reading comprehension achievement and vocabulary knowledge in descending order. However, Kang (2002) also noted that the number of studies related to strategy knowledge was limited to one study. Further, regarding cognitive skills, the self-directed ability showed a moderate effect size; however, this too was limited to one study. In the present study, almost all the studies measured cognitive ability. This indicates that additional research needs to be conducted to evaluate GO interventions' effectiveness with respect to learners' learning strategies and the self-directed ability in cognitive skills.

Furthermore, given that self-efficacy showed the highest effect size in the affective domain, the use of GOs is encouraged among students with low achievement. This is because improving the self-efficacy of students who struggle with learning would enable them to sustain their academic learning and is highly correlated with their academic achievement (Huang, 2016).

Meta-regression analysis

While several variables exhibited significant subgroup differences, only group size and frequency were identified

as significant moderators after controlling for other factors. These results are supported by previous literature and research, which revealed a correlation between higher intensive intervention dosage and improved student outcomes (Donegan & Wanzek, 2021; Gersten et al., 2020; Jung et al., 2017; Wanzek & Vaughn, 2007). However, the results of the meta-regression analysis could suggest that, if practitioners need to prioritize types of intervention dosage, they should consider organizing small group size instruction with more frequent exposure to interventions using GO. Specifically, teachers were recommended to provide GO interventions at least twice a week when planning their instruction. Regarding group size, especially for inclusive classroom teachers, they may encounter difficulties in organizing small group instruction, as it is more common for them to provide instruction at the classroom level in real education settings (An et al., 2023).

Since most previous meta-analyses examining moderator effects of GO have focused on subgroup differences, the findings of this current study may not be directly comparable to those of the previous meta-analyses (Dexter & Hughes, 2011; Guo et al., 2020; Joo, 2019; Kim et al., 2013; Liu et al., 2014; Moore & Readence, 1984; Schnotz, 2014). While previous studies have identified statistically significant subgroup differences, it is necessary to determine whether a moderator remains statistically significant after controlling for other factors. Therefore, further studies need to employ a meta-regression approach to explore multiple moderators in the model after accounting for other potential moderators, as opposed to subgroup analysis (Filderman et al., 2022; Jitendra et al., 2021).

Limitations

This study has at least two limitations. First, compared to studies involving students without disabilities, it may be difficult to generalize the results because there are fewer studies involving students with disabilities and at-risk learners. In addition, the majority of interventions were carried out in both general education classrooms and self-contained classrooms within general education schools. This pattern arises from the fact that students with learning disabilities (LDs) and at-risk learners primarily receive interventions in general classrooms, while students with intellectual disabilities (IDs) are typically placed in special education schools. In Korea, students' placements are determined with the goal of providing the most suitable environment for each individual. As a result, most at-risk learners and students with LDs find themselves in self-contained classrooms and general classrooms (Ministry of Education & Science Technology, 2008). The effectiveness of interventions using GOs (educational tools or strategies) is notably high among at-risk learners

and students with LDs within this educational placement system. Consequently, the applicability of GO-based interventions may not be consistent and generalizable across all types of learners. Future studies should delve into more specific educational contexts, such as placement and learner types, to provide a more nuanced understanding. The second limitation concerns comparing the effect sizes of each type of GO, as in many cases, the effect size segmented by GO type was closely related to the intervention subject. However, multiple types of GOs were used in most cases, and there were cases where the number of effect sizes was limited in terms of facilitating comparison by type of GO. In addition, only one study examined an intervention that exclusively uses visual display. Therefore, accumulating more studies is recommended, as is being cautious about generalizing this result in the meantime. Considering that a large effect size was observed, further studies should explore whether visual display is effective in mathematics intervention. Thus, more studies comparing the effects of various types of GOs need to be conducted.

Conclusion

This study was the first meta-analysis to include both students with and without disabilities in order to examine the effectiveness of GOs in Korea. At-risk students and students with LDs showed a large effect size in large groups, which could be a rationale for adopting GOs in an inclusive setting. Thus, teachers should be encouraged to use GOs in practice among various learner types who have academic difficulties, with particular attention to the moderators that maximize the effectiveness of interventions incorporating GOs.

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Declarations

Conflict of interest No conflict of interest declared by the authors with respect to the research, authorship, and/or publication of this article.

Ethical approval This is a meta-analysis paper which synthesized data from previous published studies in which informed consent was obtained by primary investigators will be retrieved and analyzed.

Therefore, ethics approval was not required for this meta-analysis paper.

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